

OPTIMUS PRIME / G1 EXPEDITION

THE MOUNTAIN AMPLIFIES FAILURE

A SLIP BECOMES A FALL.

A FALL BECOMES A RESCUE.

G1 must use tools, recover, and know when to stop.



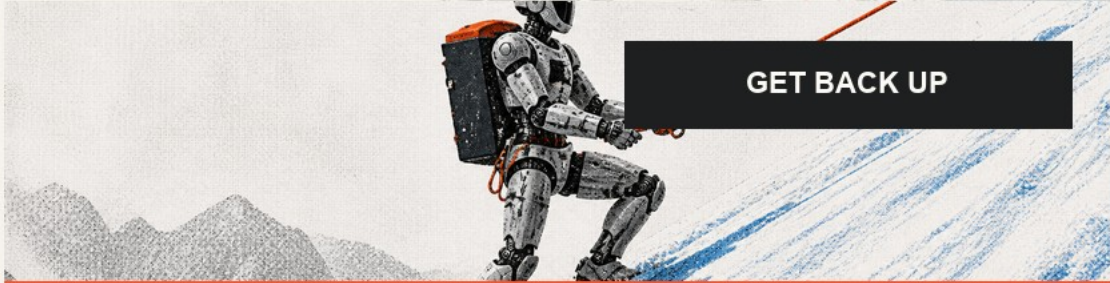
FOUR SKILLS FOR ALPINE FAILURE.

ONE EXPEDITION SYSTEM



STOP THE SLIDE

01 SELF-ARREST



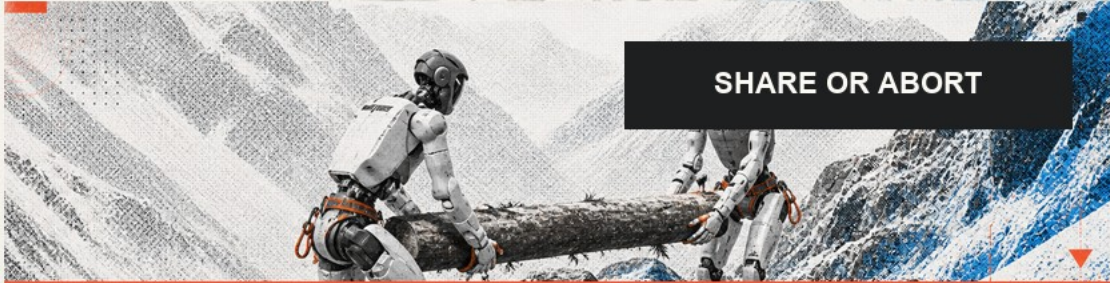
GET BACK UP

02 FALL RECOVERY



CONTROL THE DESCENT

03 CONTROLLED RAPPEL



SHARE OR ABORT

04 TEAM LIFT

TECHNICAL ARCHITECTURE

TWO SIMULATORS / ONE LIVE INFERENCE BUS

MUJOCO / SINGLE-ROBOT

SELF-ARREST · FIXED-LINE · GET-UP

PPO task policies · 256×256 MLPs
29-DoF whole-body recovery prior

ISAAC LAB / MULTI-ROBOT

COOPERATIVE LIFT

shared MAPPO · teammate attention
10 actions / G1 · frozen AGILE legs

DEPLOYED INFERENCE / REPLACES $N \times N$ ROBOT LINKS

G1 0

63 B

G1 1

63 B

G1 N

63 B

LIVEKIT ROOM

POSE · VELOCITY · LOAD

STATE FAN-OUT / STALE → ZERO

ACTOR 0

N-1 TOKENS

ACTOR 1

N-1 TOKENS

ACTOR N

N-1 TOKENS

PUBLISH ONCE / ROBOT · EVERY ACTOR RECEIVES N-1 TEAMMATE TOKENS

WHAT WE MEASURED

FIXED TESTS / PHYSICAL OUTCOMES



SELF-ARREST

9 / 9 SCENARIOS

60 / 60 unseen falls arrested



FALL RECOVERY

RECOVERS TO STAND

then continues uphill



RAPPEL

2.00 m DESCENT

7 / 10 randomized starts passed



TEAM LIFT

50 / 50 SHARE

overload or imbalance → abort

MOVEMENT / ACTION / THINKING